CUSTOMER MILESTONE BRIEF

MS #12 - Model Workcell and Context Control

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.12.0, 0.12.1, 0.12.2
Date	2026-08-11 to 2026-08-13
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #12 - Model Workcell and Context Control converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Controls model cost, context, allowed edits, retries, and evidence so AI assistance remains bounded and reproducible.

Why this matters to a customer

Controls model cost, context, allowed edits, retries, and evidence so AI assistance remains bounded and reproducible.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Added a replaceable model-provider contract with a controller-side budget boundary for calls, input/output bytes, tokens, estimated cost, and elapsed time.
- Refactored the OpenAI Responses integration behind the provider contract while retaining strict JSON Schema output, store: false, external credentials, refusal detection, and prompt-free receipts.
- Added a graph context assembler that binds approved graph roots, shared source-evidence capsules, allowed candidate files, truncation state, byte limits, and exact graph/evidence identities.
- Added an atomic constrained patch broker that validates all edits before writing and enforces allowlisted paths, text-only targets, exact matches, file size, patch size, file count, and changed-line limits.
- Added sanitized failure-analysis feedback, independently hashed model-call artifacts, run-level intelligence receipts, and Control Tower metrics for model, calls, tokens, cost, and evidence identity.
- Added a 36-fault, eight-category public calibration catalog plus an external sealed-holdout interface; public calibration is explicitly prevented from masquerading as blind evaluation.

- Added macOS/Linux and Windows model-workcell launchers, schemas, adversarial budget and atomicity tests, catalog mutation rejection tests, and full-suite verification integration.
- Added bounded Retry-After-aware exponential backoff with jitter for transient Responses API throttling and transport failures; billing, quota, refusal, schema, and other hard failures do not retry.
- Added a 25,000-token per-call output ceiling, request timeouts, conservative pre-call cost admission, safe rate-limit metadata, and prompt-free retry evidence.
- Added evaluation-wide call, token, and estimated-cost budgets plus lower per-case ceilings and configurable pacing between cases.
- Added atomic per-case checkpoints, partial stopped receipts, sanitized provider failure classification, collision-safe retry run IDs, and resumability without repeating completed cases.
- Added v1.1 evaluation receipts, checkpoint schema, cross-platform resume commands, and tests for transient 429 recovery, hard-quota rejection, budget stops, partial evidence, and resume.
- Replaced repeated full-context prompts with progressive role-specific retrieval: planners receive a compact graph and evidence catalog, and builders receive only plan-selected source capsules.
- Added fail-closed validation for planner-selected graph nodes and evidence capsule IDs plus independent 80 KB planner and builder context ceilings.
- Added exact Responses API input-token preflight before generation, a configurable per-call input ceiling, bounded count-request retries, and count identities and rate metadata in call evidence.
- Added request manifests that record prompt and payload hashes, context statistics, and selected evidence IDs without persisting provider credentials or full model prompts.
- Added an audience-safe controller-mediated transcript command that shows planner and builder artifacts while redacting verifier-private evidence by default.
- Added regression tests proving substantial context reduction, evidence-selection boundaries, exact token admission, pre-generation budget rejection, and transcript privacy.

How it differs from earlier work

MS #12 builds on MS #11 (Hardened Admitted Execution). The earlier milestone established its own bounded capability; this milestone adds model workcell and context control while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use model workcell and context control as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- This milestone establishes only the bounded capabilities and evidence listed here; later capabilities are not retroactive.
- Production readiness and platform equivalence require the explicit evidence and policy gates applicable to the customer environment.

Source of record

- CHANGELOG.md release section(s): 0.12.0, 0.12.1, 0.12.2
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity